
Program 1. : Program to hold details of 5 students and provide the facility of viewing details of the topper as well as of specific student by providing his/her roll number.

Soln:

```
#include <iostream>

using namespace std;

struct Student {
    int roll;
    string name;
    float marks;
};

int main() {
    Student s[5];
    int topperIndex = 0;
    for (int i = 0; i < 5; i++) {
        cout << "Enter Roll, Name, and Marks for student " << i + 1 << ": ";
        cin >> s[i].roll >> s[i].name >> s[i].marks;
        if (s[i].marks > s[topperIndex].marks) {
            topperIndex = i;
        }
    }
    cout << "\n--- TOPPER DETAILS ---\n";
    cout << "Roll: " << s[topperIndex].roll
        << " | Name: " << s[topperIndex].name
        << " | Marks: " << s[topperIndex].marks << endl;
    int searchRoll;
    cout << "\nEnter Roll Number to search: ";
    cin >> searchRoll;
    bool found = false;
    for (int i = 0; i < 5; i++) {
        if (s[i].roll == searchRoll) {
            cout << "Student Found -> Name: " << s[i].name
```

```
<< ", Marks: " << s[i].marks << endl;

found = true;

break;

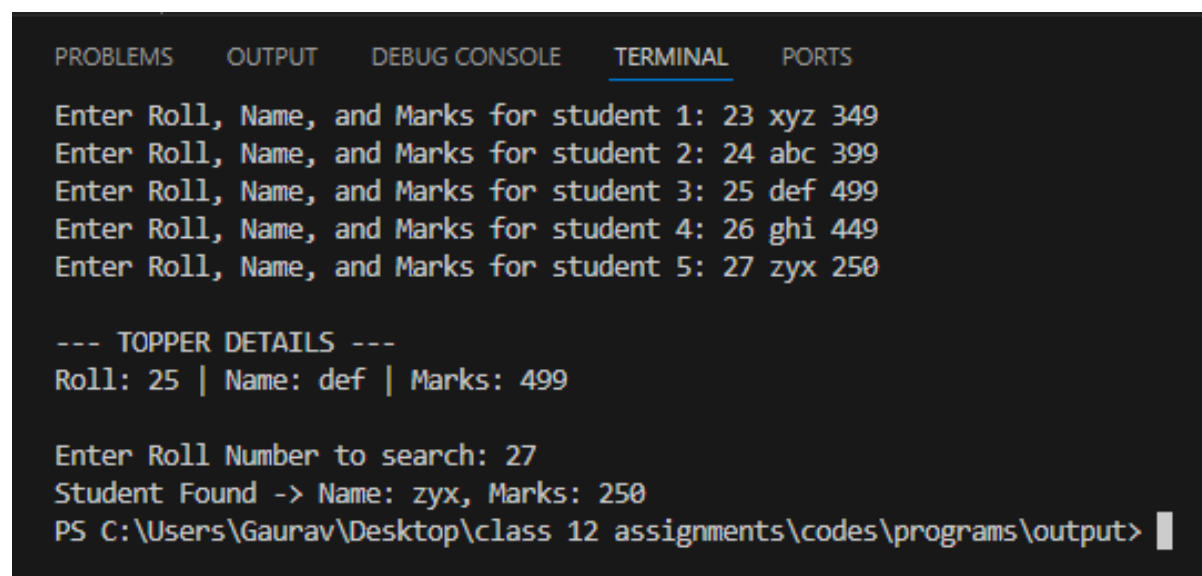
}

}

if (!found) cout << "Student with Roll Number " << searchRoll << " not found.\n";

return 0;

}
```

Output:

The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are five tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal output shows the program's execution. It starts by prompting the user to enter roll, name, and marks for five students. The input provided is: student 1: 23 xyz 349, student 2: 24 abc 399, student 3: 25 def 499, student 4: 26 ghi 449, and student 5: 27 zyx 250. After this, it displays '--- TOPPER DETAILS ---' and shows the details for the student with the highest marks: 'Roll: 25 | Name: def | Marks: 499'. Finally, it prompts for a roll number to search (27) and displays the result: 'Student Found -> Name: zyx, Marks: 250'. The prompt 'PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>' is visible at the bottom.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter Roll, Name, and Marks for student 1: 23 xyz 349
Enter Roll, Name, and Marks for student 2: 24 abc 399
Enter Roll, Name, and Marks for student 3: 25 def 499
Enter Roll, Name, and Marks for student 4: 26 ghi 449
Enter Roll, Name, and Marks for student 5: 27 zyx 250

--- TOPPER DETAILS ---
Roll: 25 | Name: def | Marks: 499

Enter Roll Number to search: 27
Student Found -> Name: zyx, Marks: 250
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>
```

Program 2. : Program to illustrate the access control in public derivation of a class with the concept of inheritance.

Soln:

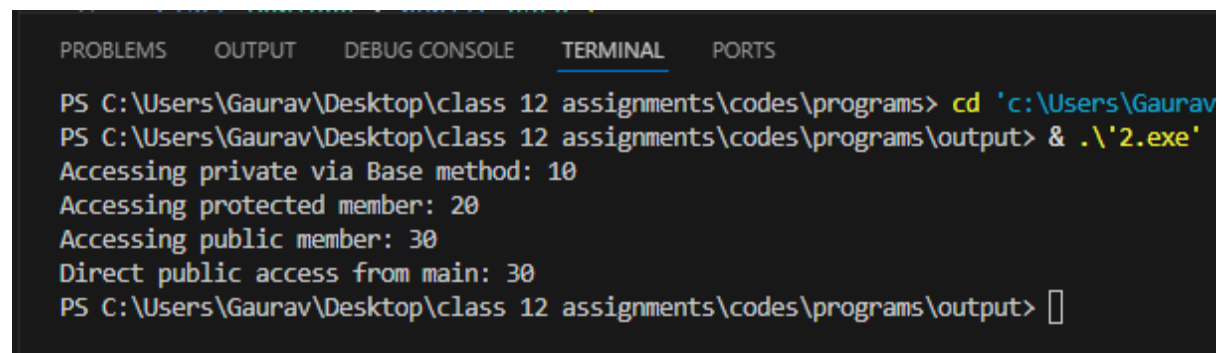
```
#include <iostream>

using namespace std;

class Base {
private:
    int priv = 10;
protected:
    int prot = 20;
public:
    int pub = 30;
    int getPrivate() { return priv; }
};

class Derived : public Base {
public:
    void display() {
        cout << "Accessing private via Base method: " << getPrivate() << endl;
        cout << "Accessing protected member: " << prot << endl;
        cout << "Accessing public member: " << pub << endl;
    }
};

int main() {
    Derived obj;
    obj.display();
    cout << "Direct public access from main: " << obj.pub << endl;
    return 0;
}
```

Output:A screenshot of a Visual Studio terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active. The command prompt shows the following sequence of commands and output:

```
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\2.exe
Accessing private via Base method: 10
Accessing protected member: 20
Accessing public member: 30
Direct public access from main: 30
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> █
```

Program 3. : Program to illustrate the use of object arrays by storing details of 5 items in an array of objects.

Soln:

```
#include <iostream>

using namespace std;

class Item {
    int id;
    string name;
    float price;
public:
    void getItem() {
        cout << "Enter Item ID, Name, and Price: ";
        cin >> id >> name >> price;
    }
    void putItem() const {
        cout << "ID: " << id << " | Name: " << name << " | Price: Rs " << price << endl;
    }
};

int main() {
    Item items[5];
    for (int i = 0; i < 5; i++) {
        cout << "Item " << i + 1 << ":\n";
        items[i].getItem();
    }
    cout << "\n--- ITEM DETAILS ---\n";
    for (int i = 0; i < 5; i++) {
        items[i].putItem();
    }
    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\3.exe
Item 1:
Enter Item ID, Name, and Price: 005 aam 99
Item 2:
Enter Item ID, Name, and Price: 006 apple 79
Item 3:
Enter Item ID, Name, and Price: 007 orange 89
Item 4:
Enter Item ID, Name, and Price: 008 pineapple 95
Item 5:
Enter Item ID, Name, and Price: 009 unknown 249

--- ITEM DETAILS ---
ID: 5 | Name: aam | Price: Rs 99
ID: 6 | Name: apple | Price: Rs 79
ID: 7 | Name: orange | Price: Rs 89
ID: 8 | Name: pineapple | Price: Rs 95
ID: 9 | Name: unknown | Price: Rs 249
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> 
```

Program 4. : Program to illustrate the working of a function returning an object.

Soln:

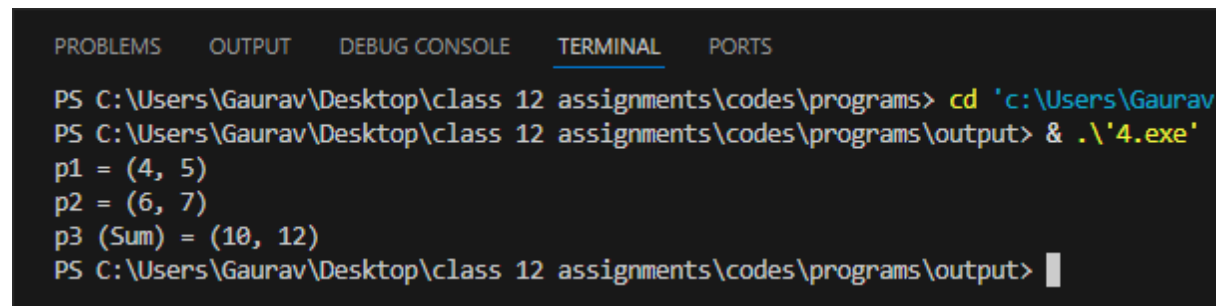
```
#include <iostream>

using namespace std;

class Point {
public:
    int x, y;
    Point(int xVal = 0, int yVal = 0) : x(xVal), y(yVal) {}
    void display() const {
        cout << "(" << x << ", " << y << ")" << endl;
    }
};

Point addPoints(Point p1, Point p2) {
    Point temp(p1.x + p2.x, p1.y + p2.y);
    return temp;
}

int main() {
    Point p1(4, 5), p2(6, 7);
    Point p3 = addPoints(p1, p2);
    cout << "p1 = "; p1.display();
    cout << "p2 = "; p2.display();
    cout << "p3 (Sum) = "; p3.display();
    return 0;
}
```

Output:A screenshot of a terminal window with a dark background. At the top, there are five tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal shows a series of commands and their output. The first command is 'cd 'c:\Users\Gaurav\...', followed by '& .\'4.exe''. The output consists of three lines: 'p1 = (4, 5)', 'p2 = (6, 7)', and 'p3 (Sum) = (10, 12)'. The prompt 'PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>' is visible at the end of the output lines.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\'4.exe'
p1 = (4, 5)
p2 = (6, 7)
p3 (Sum) = (10, 12)
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 5. : Program to illustrate the Call By Reference mechanism on objects.

Soln:

```
#include <iostream>

using namespace std;

class Number {
public:
    int val;

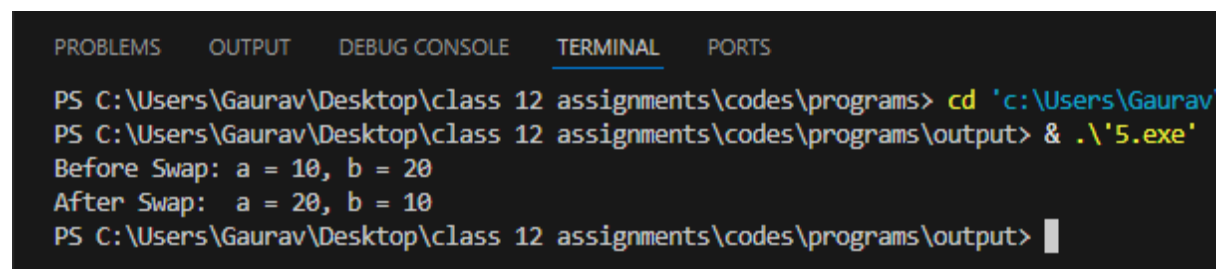
    Number(int v) : val(v) {}
};

void swapObjects(Number &n1, Number &n2) {
    int temp = n1.val;
    n1.val = n2.val;
    n2.val = temp;
}

int main() {
    Number a(10), b(20);

    cout << "Before Swap: a = " << a.val << ", b = " << b.val << endl;
    swapObjects(a, b);
    cout << "After Swap: a = " << a.val << ", b = " << b.val << endl;
    return 0;
}
```

Output :



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\5.exe
Before Swap: a = 10, b = 20
After Swap: a = 20, b = 10
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> █
```

Program 6. : Program to Sort elements of an array.**Soln:**

```
#include <iostream>

using namespace std;

int main() {

    int n, arr[100];

    cout << "Enter number of elements: ";

    cin >> n;

    cout << "Enter " << n << " integers: ";

    for (int i = 0; i < n; i++) cin >> arr[i];

    for (int i = 0; i < n - 1; i++) {

        for (int j = 0; j < n - i - 1; j++) {

            if (arr[j] > arr[j + 1]) {

                int temp = arr[j];

                arr[j] = arr[j + 1];

                arr[j + 1] = temp;

            }

        }

    }

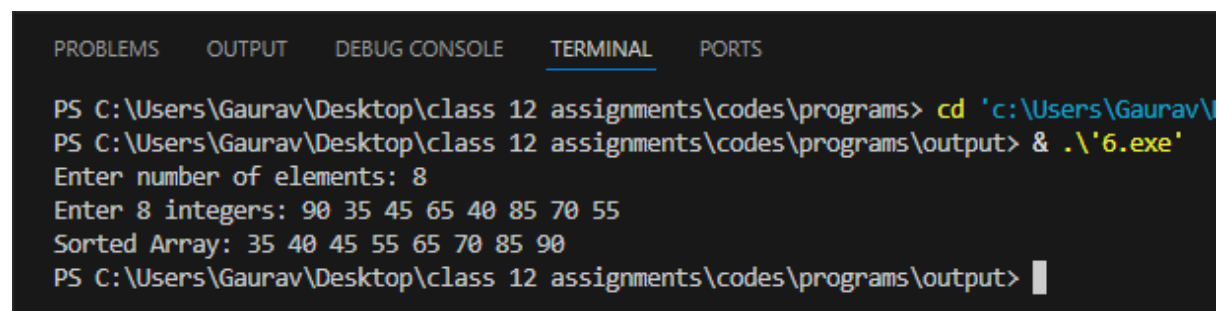
    cout << "Sorted Array: ";

    for (int i = 0; i < n; i++) cout << arr[i] << " ";

    cout << endl;

    return 0;

}
```

Output:A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal shows the following text: 'PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output' & .\'6.exe'', 'Enter number of elements: 8', 'Enter 8 integers: 90 35 45 65 40 85 70 55', 'Sorted Array: 35 40 45 55 65 70 85 90', and 'PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>'.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output' & .\'6.exe'
Enter number of elements: 8
Enter 8 integers: 90 35 45 65 40 85 70 55
Sorted Array: 35 40 45 55 65 70 85 90
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 7. : Program demonstrating Linear searching.

Soln:

```
#include <iostream>

using namespace std;

int main() {

    int n, key, arr[100];

    cout << "Enter array size: ";

    cin >> n;

    cout << "Enter elements: ";

    for (int i = 0; i < n; i++) cin >> arr[i];

    cout << "Enter element to search: ";

    cin >> key;

    int pos = -1;

    for (int i = 0; i < n; i++) {

        if (arr[i] == key) {

            pos = i;

            break;

        }

    }

    if (pos != -1)

        cout << "Element found at index " << pos << " (Position " << pos + 1 << ")\n";

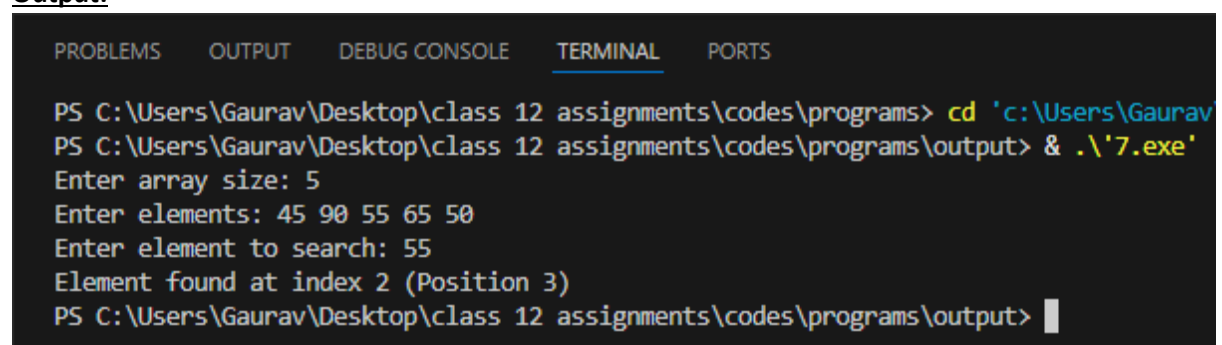
    else

        cout << "Element not found.\n";

    return 0;

}
```

Output:

A screenshot of a Windows terminal window showing the execution of a C++ program. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The command prompt shows the user navigating to the directory 'C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs' and running the program '7.exe'. The program prompts for array size (5), elements (45 90 55 65 50), and the element to search (55). It then outputs 'Element found at index 2 (Position 3)'.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\7.exe
Enter array size: 5
Enter elements: 45 90 55 65 50
Enter element to search: 55
Element found at index 2 (Position 3)
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> █
```

Program 8. : Program demonstrating Binary search.**Soln:**

```
#include <iostream>

using namespace std;

int main() {
    int n, key, arr[100];

    cout << "Enter size of sorted array: ";

    cin >> n;

    cout << "Enter sorted elements: ";
    for (int i = 0; i < n; i++) cin >> arr[i];

    cout << "Enter element to search: ";

    cin >> key;

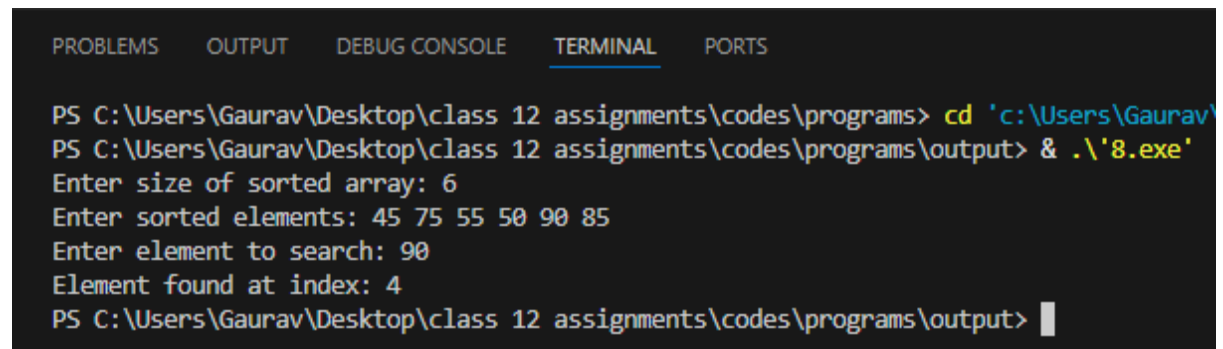
    int low = 0, high = n - 1, mid, index = -1;

    while (low <= high) {
        mid = low + (high - low) / 2;

        if (arr[mid] == key) {
            index = mid;
            break;
        } else if (arr[mid] < key) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }

    if (index != -1)
        cout << "Element found at index: " << index << endl;
    else
        cout << "Element not found.\n";

    return 0;
}
```

Output:

The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are five tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal content shows a series of commands and user inputs. The first command is `cd 'c:\Users\Gaurav\`, followed by `PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\'8.exe'`. The program then prompts for input: 'Enter size of sorted array: 6', 'Enter sorted elements: 45 75 55 50 90 85', and 'Enter element to search: 90'. The output of the program is 'Element found at index: 4'. The final line shows the prompt `PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>` with a cursor.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\'8.exe'
Enter size of sorted array: 6
Enter sorted elements: 45 75 55 50 90 85
Enter element to search: 90
Element found at index: 4
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 9. : Program for Inserting elements into the array.

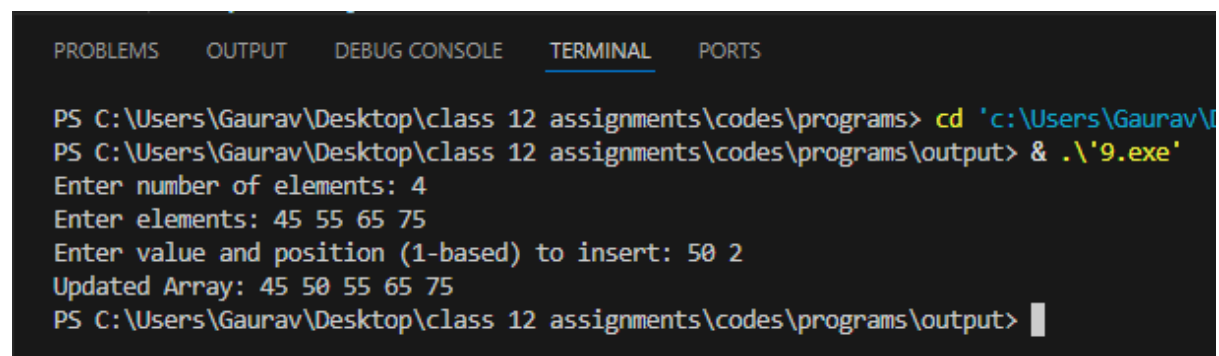
Soln:

```
#include <iostream>

using namespace std;

int main() {
    int n, val, pos, arr[100];
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter elements: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    cout << "Enter value and position (1-based) to insert: ";
    cin >> val >> pos;
    for (int i = n; i >= pos; i--) {
        arr[i] = arr[i - 1];
    }
    arr[pos - 1] = val;
    n++;
    cout << "Updated Array: ";
    for (int i = 0; i < n; i++) cout << arr[i] << " ";
    cout << endl;
    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\9.exe
Enter number of elements: 4
Enter elements: 45 55 65 75
Enter value and position (1-based) to insert: 50 2
Updated Array: 45 50 55 65 75
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

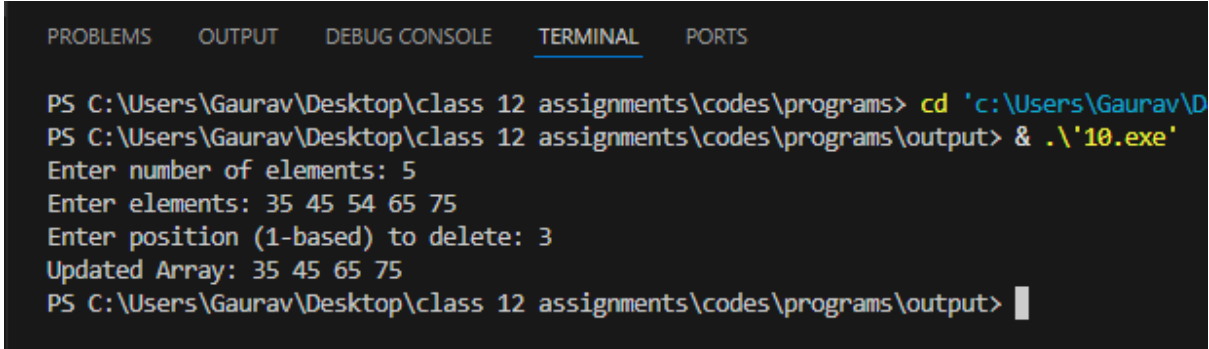
Program 10. : Program for Deleting elements on the array.

Soln:

```
#include <iostream>

using namespace std;

int main() {
    int n, pos, arr[100];
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter elements: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    cout << "Enter position (1-based) to delete: ";
    cin >> pos;
    if (pos < 1 || pos > n) {
        cout << "Invalid position!\n";
    } else {
        for (int i = pos - 1; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }
        n--;
        cout << "Updated Array: ";
        for (int i = 0; i < n; i++) cout << arr[i] << " ";
        cout << endl;
    }
    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\'10.exe'
Enter number of elements: 5
Enter elements: 35 45 54 65 75
Enter position (1-based) to delete: 3
Updated Array: 35 45 65 75
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 11. : Write a C++ program that inputs a line of text and a substring to search for. Then it should display the position of the first and last occurrence of the word.

Soln:

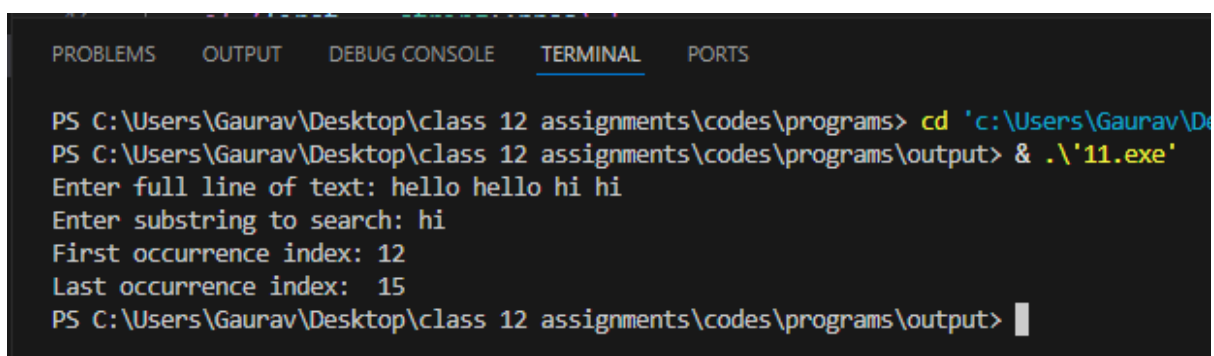
```
#include <iostream>

#include <string>

using namespace std;

int main() {
    string text, sub;
    cout << "Enter full line of text: ";
    getline(cin, text);
    cout << "Enter substring to search: ";
    cin >> sub;
    size_t first = text.find(sub);
    size_t last = text.rfind(sub);
    if (first == string::npos) {
        cout << "Substring not found in the given text.\n";
    } else {
        cout << "First occurrence index: " << first << endl;
        cout << "Last occurrence index: " << last << endl;
    }
    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\11.exe
Enter full line of text: hello hello hi hi
Enter substring to search: hi
First occurrence index: 12
Last occurrence index: 15
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 12. : Program to illustrate summation of Matrices.

Soln:

```
#include <iostream>

using namespace std;

int main() {

    int r, c, a[10][10], b[10][10], sum[10][10];

    cout << "Enter rows and columns: ";

    cin >> r >> c;

    cout << "Enter Matrix A elements:\n";

    for (int i = 0; i < r; i++)

        for (int j = 0; j < c; j++)

            cin >> a[i][j];

    cout << "Enter Matrix B elements:\n";

    for (int i = 0; i < r; i++)

        for (int j = 0; j < c; j++)

            cin >> b[i][j];

    for (int i = 0; i < r; i++)

        for (int j = 0; j < c; j++)

            sum[i][j] = a[i][j] + b[i][j];

    cout << "\nResultant Sum Matrix:\n";

    for (int i = 0; i < r; i++) {

        for (int j = 0; j < c; j++) {

            cout << sum[i][j] << " ";

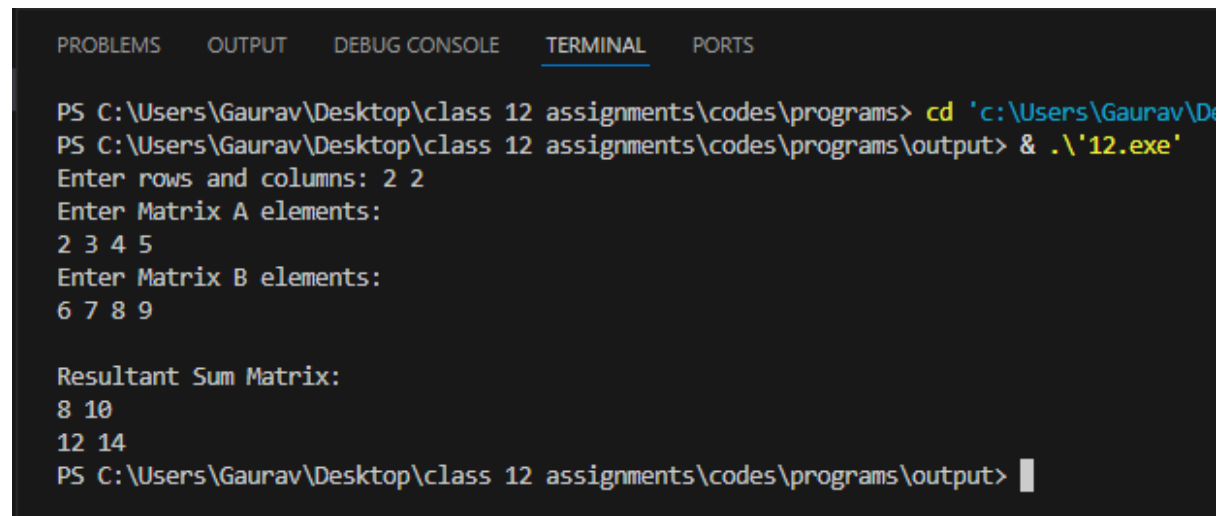
        }

        cout << endl;

    }

    return 0;

}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\12.exe
Enter rows and columns: 2 2
Enter Matrix A elements:
2 3 4 5
Enter Matrix B elements:
6 7 8 9

Resultant Sum Matrix:
8 10
12 14
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 13. : Write a C++ program to find the Longest consecutive sequence of integers with no primes. The program should accept a number N and print out the Largest block of integers between 2 and N with no primes.

Soln:

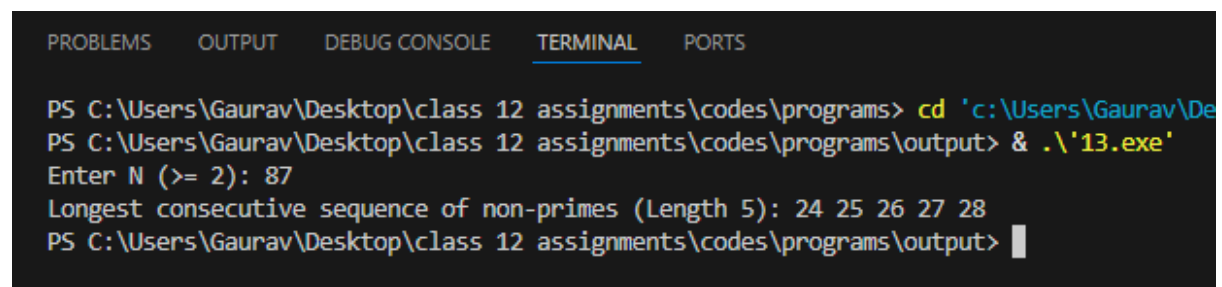
```
#include <iostream>

using namespace std;

bool isPrime(int n) {
    if (n < 2) return false;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) return false;
    }
    return true;
}

int main() {
    int n;
    cout << "Enter N (>= 2): ";
    cin >> n;
    int maxLen = 0, maxStart = 0;
    int curLen = 0, curStart = 0;
    for (int i = 2; i <= n; i++) {
        if (!isPrime(i)) {
            if (curLen == 0) curStart = i;
            curLen++;
            if (curLen > maxLen) {
                maxLen = curLen;
                maxStart = curStart;
            } else {
                curLen = 0;
            }
        }
    }
    if (maxLen > 0) {
```

```
cout << "Longest consecutive sequence of non-primes (Length " << maxLen << "): ";  
for (int i = maxStart; i < maxStart + maxLen; i++) {  
    cout << i << " ";  
}  
cout << endl;  
} else {  
    cout << "No composite sequence found between 2 and " << n << ".\n";  
}  
return 0;  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\'13.exe'  
Enter N (>= 2): 87  
Longest consecutive sequence of non-primes (Length 5): 24 25 26 27 28  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 14. : Write a function in C++ to count the words "this" and "these" present in a text file "ARTICLE.TXT".

Soln:

```
#include <iostream>

#include <fstream>

#include <string>

#include <algorithm>

using namespace std;

void countWords() {

    ifstream file("ARTICLE.TXT");

    if (!file) {

        cout << "Error: Cannot open ARTICLE.TXT\n";

        return;

    }

    string word;

    int countThis = 0, countThese = 0;

    while (file >> word) {

        string cleanWord = "";

        for (char ch : word) {

            if (isalpha(ch)) cleanWord += tolower(ch);

        }

        if (cleanWord == "this") countThis++;

        if (cleanWord == "these") countThese++;

    }

    file.close();

    cout << "Count of 'this': " << countThis << endl;

    cout << "Count of 'these': " << countThese << endl;

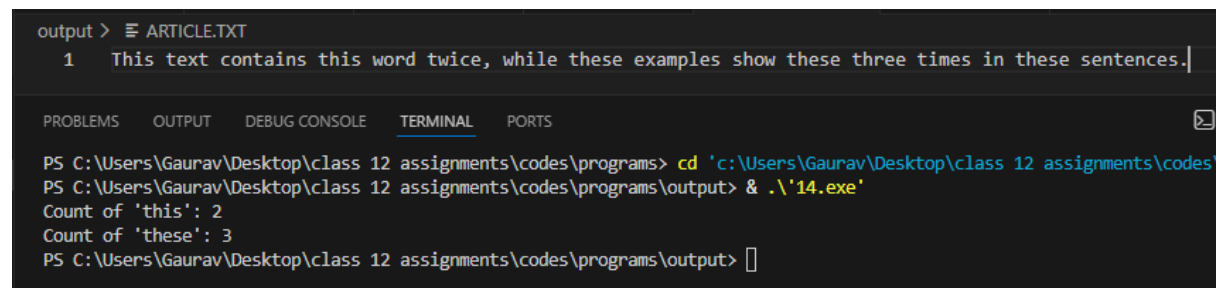
}

int main() {

    countWords();

}
```

```
return 0;  
}
```

Output:

```
output > ARTICLE.TXT  
1 This text contains this word twice, while these examples show these three times in these sentences.  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\14.exe  
Count of 'this': 2  
Count of 'these': 3  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>
```

Program 15. : Write a C++ program that displays the size of a file in bytes.

Soln:

```
#include <iostream>

#include <fstream>

#include <string>

using namespace std;

int main() {

    string filename;

    cout << "Enter file name: ";

    cin >> filename;

    ifstream file(filename, ios::binary | ios::ate);

    if (!file) {

        cout << "Error opening file or file does not exist!\n";

        return 1;

    }

    streampos size = file.tellg();

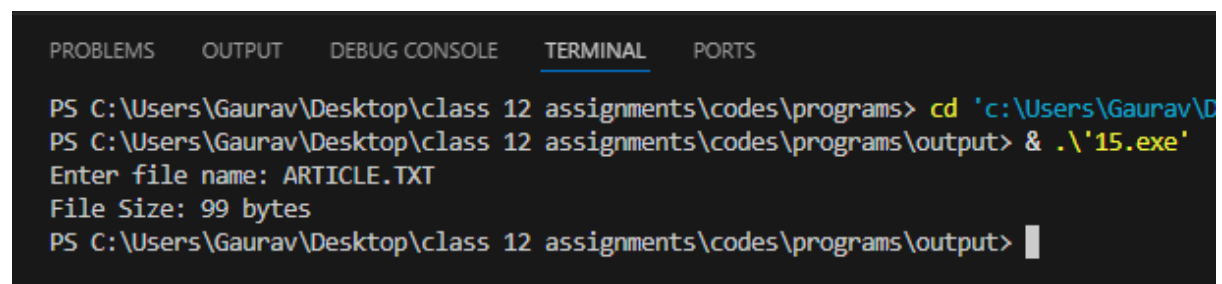
    file.close();

    cout << "File Size: " << size << " bytes\n";

    return 0;

}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\15.exe
Enter file name: ARTICLE.TXT
File Size: 99 bytes
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 16. : Program for Insertion at the beginning of a List.

Soln:

```
#include <iostream>

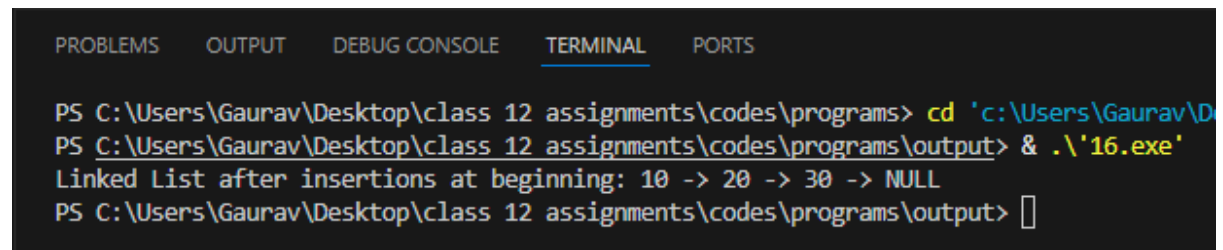
using namespace std;

struct Node {
    int data;
    Node* next;
};

void insertAtBeginning(Node*& head, int val) {
    Node* newNode = new Node{val, head};
    head = newNode;
}

void display(Node* head) {
    Node* temp = head;
    while (temp != nullptr) {
        cout << temp->data << " -> ";
        temp = temp->next;
    }
    cout << "NULL\n";
}

int main() {
    Node* head = nullptr;
    insertAtBeginning(head, 30);
    insertAtBeginning(head, 20);
    insertAtBeginning(head, 10);
    cout << "Linked List after insertions at beginning: ";
    display(head);
    return 0;
}
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output' & .\'16.exe'
Linked List after insertions at beginning: 10 -> 20 -> 30 -> NULL
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> 
```

Program 17. : Program for Insertion at the end of a List.

Soln:

```
#include <iostream>

using namespace std;

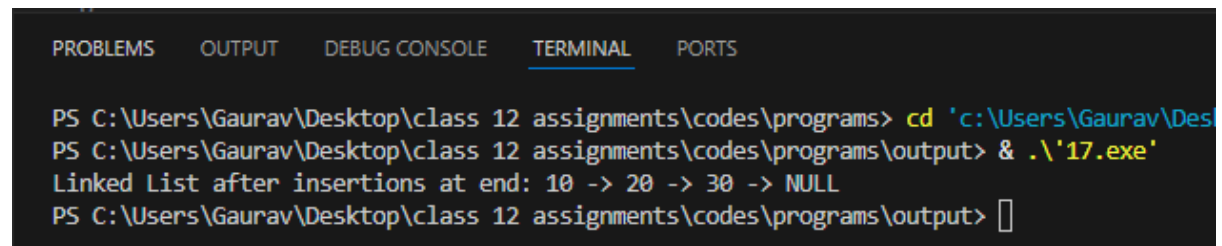
struct Node {
    int data;
    Node* next;
};

void insertAtEnd(Node*& head, int val) {
    Node* newNode = new Node{val, nullptr};
    if (head == nullptr) {
        head = newNode;
        return;
    }
    Node* temp = head;
    while (temp->next != nullptr) {
        temp = temp->next;
    }
    temp->next = newNode;
}

void display(Node* head) {
    Node* temp = head;
    while (temp != nullptr) {
        cout << temp->data << " -> ";
        temp = temp->next;
    }
    cout << "NULL\n";
}

int main() {
    Node* head = nullptr;
```

```
insertAtEnd(head, 10);  
insertAtEnd(head, 20);  
insertAtEnd(head, 30);  
cout << "Linked List after insertions at end: ";  
display(head);  
return 0;  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output'  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\17.exe  
Linked List after insertions at end: 10 -> 20 -> 30 -> NULL  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> █
```

Program 18. : Write a C++ program to declare a pointer to an integer, assign a value to it, and print the value using the pointer.

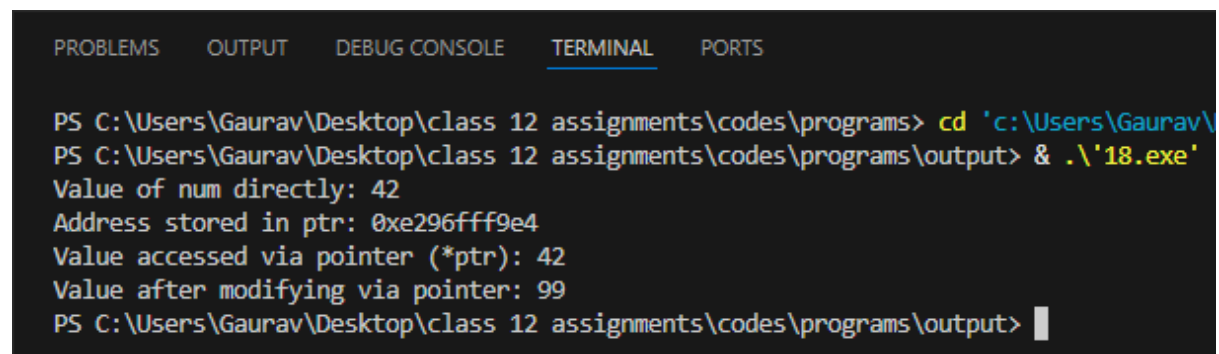
Soln:

```
#include <iostream>

using namespace std;

int main() {
    int num = 42;
    int* ptr = &num;
    cout << "Value of num directly: " << num << endl;
    cout << "Address stored in ptr: " << ptr << endl;
    cout << "Value accessed via pointer (*ptr): " << *ptr << endl;
    *ptr = 99;
    cout << "Value after modifying via pointer: " << num << endl;
    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output' & .\'18.exe'
Value of num directly: 42
Address stored in ptr: 0xe296fff9e4
Value accessed via pointer (*ptr): 42
Value after modifying via pointer: 99
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>
```

Program 19. : Write a C++ program that demonstrates polymorphism. Create a base class Shape with a function area(). Derive a class Circle and override the area() function to calculate the area of a circle.

Soln:

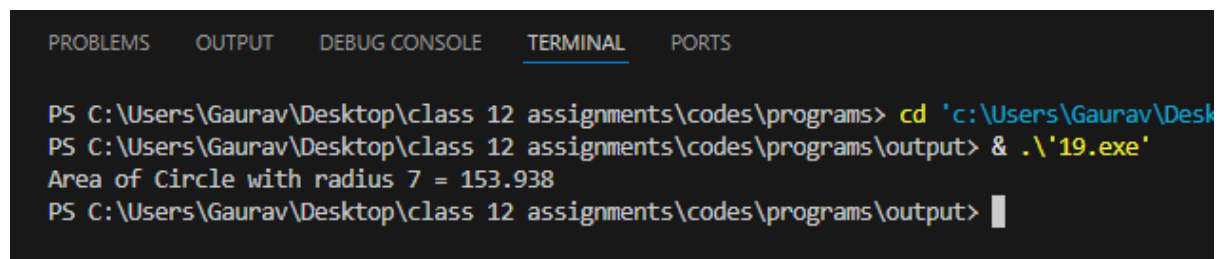
```
#include <iostream>

using namespace std;

class Shape {
public:
    virtual void area() const {
        cout << "Base Shape area undefined.\n";
    }
    virtual ~Shape() {}
};

class Circle : public Shape {
    double radius;
public:
    Circle(double r) : radius(r) {}
    void area() const override {
        cout << "Area of Circle with radius " << radius
            << " = " << 3.14159 * radius * radius << endl;
    }
};

int main() {
    Shape* shapePtr;
    Circle c(7.0);
    shapePtr = &c;
    shapePtr->area(); // Dynamic dispatch
    return 0;
}
```

Output:

The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are five tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal content shows a PowerShell prompt 'PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs>' followed by the command 'cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output' & .\'19.exe''. The output of the command is 'Area of Circle with radius 7 = 153.938'. The prompt then moves to the next line: 'PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output>' with a cursor at the end.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output' & .\'19.exe'
Area of Circle with radius 7 = 153.938
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 20. : Write a C++ program with a class Book that uses a constructor to initialize title and author, and a destructor that prints a message when an object is destroyed.

Soln:

```
#include <iostream>

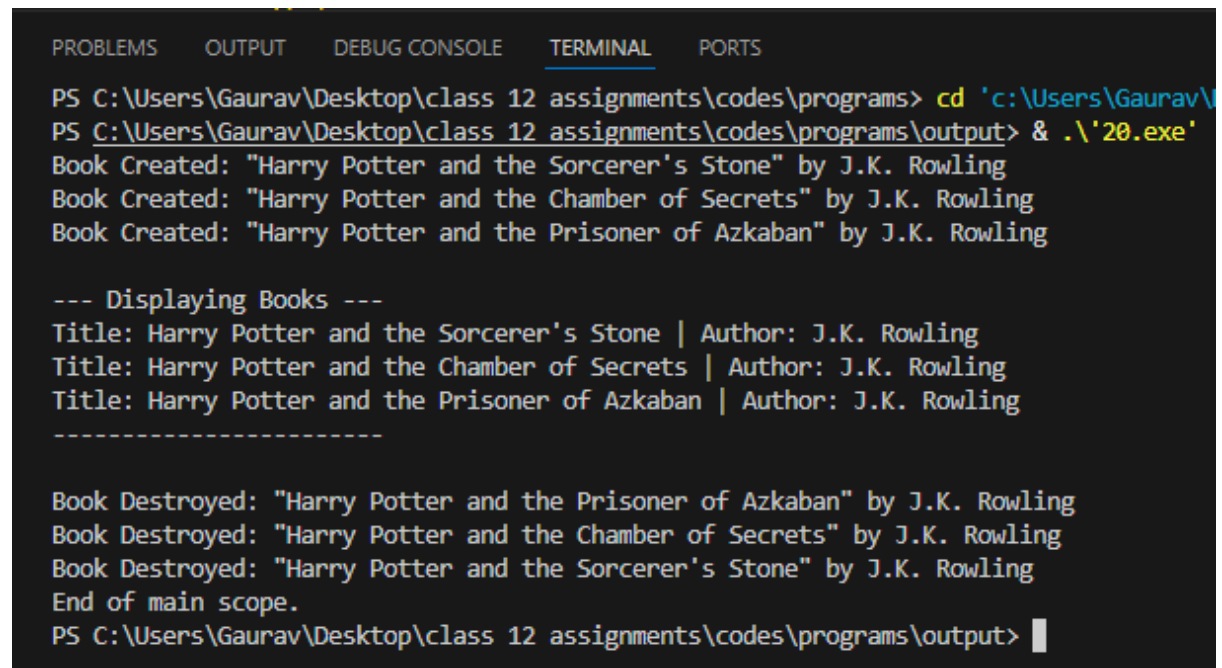
using namespace std;

class Book {
    string title;
    string author;
public:
    Book(string t, string a) : title(t), author(a) {
        cout << "Book Created: \"\" << title << \"\" by \"\" << author << endl;
    }
    ~Book() {
        cout << "Book Destroyed: \"\" << title << \"\" by \"\" << author << endl;
    }
    void display() const {
        cout << "Title: \"\" << title << \" | Author: \"\" << author << endl;
    }
};

int main() {
    {
        Book b1("Harry Potter and the Sorcerer's Stone", "J.K. Rowling");
        Book b2("Harry Potter and the Chamber of Secrets", "J.K. Rowling");
        Book b3("Harry Potter and the Prisoner of Azkaban", "J.K. Rowling");
        cout << "\\n--- Displaying Books ---\\n";
        b1.display();
        b2.display();
        b3.display();
        cout << "-----\\n\\n";
    }
}
```



```
cout << "End of main scope.\n";  
return 0;  
}
```

Output:A screenshot of a C++ program's output in a terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The command prompt shows the user navigating to the directory 'C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs' and running the program '20.exe'. The output displays three book creation messages, a separator line, three book display messages, another separator line, three book destruction messages, and finally 'End of main scope.'.

```
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output' & .\20.exe  
Book Created: "Harry Potter and the Sorcerer's Stone" by J.K. Rowling  
Book Created: "Harry Potter and the Chamber of Secrets" by J.K. Rowling  
Book Created: "Harry Potter and the Prisoner of Azkaban" by J.K. Rowling  
  
--- Displaying Books ---  
Title: Harry Potter and the Sorcerer's Stone | Author: J.K. Rowling  
Title: Harry Potter and the Chamber of Secrets | Author: J.K. Rowling  
Title: Harry Potter and the Prisoner of Azkaban | Author: J.K. Rowling  
-----  
  
Book Destroyed: "Harry Potter and the Prisoner of Azkaban" by J.K. Rowling  
Book Destroyed: "Harry Potter and the Chamber of Secrets" by J.K. Rowling  
Book Destroyed: "Harry Potter and the Sorcerer's Stone" by J.K. Rowling  
End of main scope.  
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 21. : Write a C++ program to create a binary file for records (ID, Name, Marks), a text file for logs, update records in the binary file, append data to the text file, and query records by ID with proper error handling using fstream.

Soln:

```
#include <iostream>

#include <fstream>

#include <cstring>

using namespace std;

struct Record {

    int id;

    char name[50];

    float marks;

};

void logMessage(const string& msg) {

    ofstream logFile("log.txt", ios::app);

    if (logFile) {

        logFile << msg << endl;

    }

}

void addRecord(int id, const char* name, float marks) {

    ofstream out("records.dat", ios::binary | ios::app);

    if (!out) {

        cout << "Error opening binary file!\n";

        return;

    }

    Record r;

    r.id = id;

    strncpy(r.name, name, sizeof(r.name) - 1);

    r.name[sizeof(r.name) - 1] = '\0';

    r.marks = marks;

    out.write(reinterpret_cast<char*>(&r), sizeof(Record));

}
```

```
    out.close();

    logMessage("Added record with ID: " + to_string(id));
}

void queryRecord(int id) {
    ifstream in("records.dat", ios::binary);

    if (!in) {
        cout << "Binary file not found!\n";
        return;
    }

    Record r;

    bool found = false;

    while (in.read(reinterpret_cast<char*>(&r), sizeof(Record))) {
        if (r.id == id) {
            cout << "Found -> ID: " << r.id << ", Name: " << r.name << ", Marks: " << r.marks << endl;

            found = true;
            break;
        }
    }

    if (!found) cout << "Record with ID " << id << " not found.\n";

    in.close();
}

void updateMarks(int id, float newMarks) {
    fstream file("records.dat", ios::binary | ios::in | ios::out);

    if (!file) {
        cout << "File error!\n";
        return;
    }

    Record r;

    bool found = false;

    while (file.read(reinterpret_cast<char*>(&r), sizeof(Record))) {
```

```
    if (r.id == id) {
        r.marks = newMarks;
        file.seekp(-static_cast<streamoff>(sizeof(Record)), ios::cur);
        file.write(reinterpret_cast<char*>(&r), sizeof(Record));
        found = true;
        logMessage("Updated ID " + to_string(id) + " with new marks: " +
to_string(newMarks));
        cout << "Record updated.\n";
        break;
    }
}
if (!found) cout << "ID not found for update.\n";
file.close();
}

int main() {
    addRecord(101, "Alice", 88.5);
    addRecord(102, "Bob", 92.0);
    cout << "Querying ID 101:\n";
    queryRecord(101);
    cout << "\nUpdating Marks for ID 101:\n";
    updateMarks(101, 95.0);
    cout << "\nQuerying ID 101 after update:\n";
    queryRecord(101);
    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs> cd 'c:\Users\Gaurav\D
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\'21.exe'
Querying ID 101:
Found -> ID: 101, Name: Alice, Marks: 88.5

Updating Marks for ID 101:
Record updated.

Querying ID 101 after update:
Found -> ID: 101, Name: Alice, Marks: 95
PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> |
```

Program 22. : Write a menu-driven C++ program to implement stack operations (push, pop, peek, and display) using a linear stack with arrays, a circular stack with arrays, and a stack with a linked list.

Soln:

```
#include <iostream>

using namespace std;

// 1. Linear Stack using Fixed Array

class LinearStack {

    int top, cap;

    int arr[50];

public:

    LinearStack(int c = 10) : top(-1), cap(c) {}

    void push(int val) {

        if (top == cap - 1) {

            cout << "Stack Overflow!\n";

            return;

        }

        arr[++top] = val;

        cout << "Pushed: " << val << endl;

    }

    void pop() {

        if (top == -1) {

            cout << "Stack Underflow!\n";

            return;

        }

        cout << "Popped: " << arr[top--] << endl;

    }

    void peek() {

        if (top == -1) {

            cout << "Stack is Empty.\n";

        } else {
```

```
        cout << "Top element: " << arr[top] << endl;
    }
}

void display() {
    if (top == -1) {
        cout << "Stack is Empty.\n";
        return;
    }
    cout << "Linear Stack (Top to Bottom): ";
    for (int i = top; i >= 0; i--) {
        cout << arr[i] << " ";
    }
    cout << endl;
}
};

// 2. Circular Stack using Array (With Wrap-around Logic)
class CircularStack {
    int top, cap, size;
    int arr[50];
public:
    CircularStack(int c = 10) : top(-1), cap(c), size(0) {}
    void push(int val) {
        if (size == cap) {
            cout << "Circular Stack Overflow!\n";
            return;
        }
        top = (top + 1) % cap;
        arr[top] = val;
        size++;
        cout << "Pushed: " << val << endl;
```

```
}  
  
void pop() {  
    if (size == 0) {  
        cout << "Circular Stack Underflow!\n";  
        return;  
    }  
  
    cout << "Popped: " << arr[top] << endl;  
    top = (top - 1 + cap) % cap;  
    size--;  
}  
  
void peek() {  
    if (size == 0) {  
        cout << "Circular Stack is Empty.\n";  
    } else {  
        cout << "Top element: " << arr[top] << endl;  
    }  
}  
  
void display() {  
    if (size == 0) {  
        cout << "Circular Stack is Empty.\n";  
        return;  
    }  
  
    cout << "Circular Stack (Top to Bottom): ";  
    int curr = top;  
    for (int i = 0; i < size; i++) {  
        cout << arr[curr] << " ";  
        curr = (curr - 1 + cap) % cap;  
    }  
  
    cout << endl;  
}
```



```
};

// 3. Stack using Linked List

struct SNode {
    int data;
    SNode* next;
};

class LinkedStack {
    SNode* top;
public:
    LinkedStack() : top(nullptr) {}
    void push(int val) {
        SNode* newNode = new SNode();
        newNode->data = val;
        newNode->next = top;
        top = newNode;
        cout << "Pushed: " << val << endl;
    }
    void pop() {
        if (!top) {
            cout << "Stack Underflow!\n";
            return;
        }
        SNode* temp = top;
        cout << "Popped: " << temp->data << endl;
        top = top->next;
        delete temp;
    }
    void peek() {
        if (!top) {
            cout << "Stack is Empty.\n";
```

```
    } else {
        cout << "Top element: " << top->data << endl;
    }
}

void display() {
    if (!top) {
        cout << "Stack is Empty.\n";
        return;
    }
    cout << "Linked Stack (Top to Bottom): ";
    for (SNode* p = top; p != nullptr; p = p->next) {
        cout << p->data << " -> ";
    }
    cout << "NULL\n";
}

// Destructor to clean up memory
~LinkedList() {
    while (top != nullptr) {
        SNode* temp = top;
        top = top->next;
        delete temp;
    }
}

};

int main() {
    LinearStack ls;
    CircularStack cs;
    LinkedList lks;
    int type, op, val;
    while (true) {
        cout << "\n=====
```

```
cout << "    MAIN MENU    \n";
cout << "===== \n";
cout << "1. Linear Stack (Array)\n";
cout << "2. Circular Stack (Array)\n";
cout << "3. Linked List Stack\n";
cout << "4. Exit Program\n";
cout << "Enter your choice (1-4): ";
cin >> type;
if (type == 4) break;
if (type < 1 || type > 4) {
    cout << "Invalid choice! Please select 1-4.\n";
    continue;
}
while (true) {
    cout << "\n--- Operations Sub-Menu ---\n";
    cout << "1. Push\n2. Pop\n3. Peek\n4. Display\n5. Go back to Main Menu\n";
    cout << "Enter operation (1-5): ";
    cin >> op;
    if (op == 5) break;
    if (op == 1) {
        cout << "Enter value to push: ";
        cin >> val;
    }
    // Route to correct object based on selected main type
    if (type == 1) {
        if (op == 1) ls.push(val);
        else if (op == 2) ls.pop();
        else if (op == 3) ls.peek();
        else if (op == 4) ls.display();
        else cout << "Invalid operation!\n";
    }
}
```

```
    }  
    else if (type == 2) {  
        if (op == 1) cs.push(val);  
        else if (op == 2) cs.pop();  
        else if (op == 3) cs.peak();  
        else if (op == 4) cs.display();  
        else cout << "Invalid operation!\n";  
    }  
    else if (type == 3) {  
        if (op == 1) lks.push(val);  
        else if (op == 2) lks.pop();  
        else if (op == 3) lks.peak();  
        else if (op == 4) lks.display();  
        else cout << "Invalid operation!\n";  
    }  
    }  
    }  
    cout << "\nProgram exited successfully.\n";  
    return 0;  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Gaurav\Desktop\class 12 assignments\codes\programs\output> & .\'22.exe'

=====
                MAIN MENU
=====
1. Linear Stack (Array)
2. Circular Stack (Array)
3. Linked List Stack
4. Exit Program
Enter your choice (1-4): 3

--- Operations Sub-Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Go back to Main Menu
Enter operation (1-5): 1
Enter value to push: 50
Pushed: 50

--- Operations Sub-Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Go back to Main Menu
Enter operation (1-5): 1
Enter value to push: 100
Pushed: 100

--- Operations Sub-Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Go back to Main Menu
Enter operation (1-5): 4
Linked Stack (Top to Bottom): 100 -> 50 -> NULL
```